



Algebraic expressions

Teacher resource

Question 1

Which expression is the same as $4m^2$?

- ☐ $8 \times m \times m$
- ☐ $4 \times m \times m$
- ☐ $4 + m + m$
- ☐ $4 \times 4 \times m \times m$
- ☐ $4 + m \times m$

Answer: B

Skill: Students interpret simple algebraic expressions.

Background information

Students should understand that a variable stands for a number and not an object. For example, if a is the number of apples, $5a$ represents 5 times as many apples as 'a apples' not five apples.

They should also understand that when dealing with variables, the properties of numbers and operations, as well as the links between operations, still apply.

Additional questions

1. Explain why $(5m)^2$ has the same value as $25m^2$.
2. Justify that the equation $6m^2 = 3m \times 2m$ is true.

Question 2

Sally added 15 to a positive number, p , and then multiplied the result by 3. She would get the same result if she:

- ☐ multiplied p by 3, then added 15
- ☐ multiplied p by 15, then added 3
- ☐ added 18 to p , then multiplied by 3
- ☐ multiplied p by 3, then added 45

Answer: D

Skill: Students apply links between operations when using variables.

Additional questions

1. Express your answer as an equation.
2. Write an expression that is equivalent to $k(k - 6)$.
3. Write an expression that is equivalent to $m^2 + 7m$.

Question 3

Lim thought of a first number. She added 4, then doubled the result and then subtracted 8 to produce a second number.

If m represents Lim's first number, which expression represents her second number?

- ☐ $(2m + 4) - 8$
- ☐ $2(m - 8) + 4$
- ☐ $2(m + 4) - 8$
- ☐ $2(m + 4 - 8)$

Answer: C

Skill: Students represent a situation using variables

Additional questions

- Peter thought of a number.
He added 5, tripled the result and then subtracted 11.
Represent this information as an algebraic expression.
- The number Peter calculated by using his process was 10.
What number did Peter first think of?

Question 4

Bob was asked to simplify some algebraic expressions.
Choose those that Bob did **not** simplify correctly.
There may be more than one incorrect answer.

	Expression	Bob's simplified expression
☐	$p \times p \div p$	p
☐	$p - p$	0
☐	$2p \div 2p$	1
☐	$3(2p)^2$	$6p^2$

Answer: D

Skill: Students apply properties of operations when using variables.

Additional questions

Complete the equations below to make each true.

- $6p = p \times 2 \times ?$
- $7p \times p \div ? = 7$
- $5 = ? \div 2p$
- $p - ? + 1 = p - 2$

Student activity: Algebraic expressions

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